Dynamic OSM-Based Profile Routing from Public GPS Trace Signals

Anonymous Author(s)

Abstract

Personalized route recommendation is usually evaluated with clean navigation logs, explicit user histories, or proprietary trajectory data. This paper studies a weaker but practical setting: route ranking from OpenStreetMap (OSM) data and public GPS-derived movement signals. We present a working route-ranking system that generates OSM route candidates, extracts interpretable route features, constructs contextual pseudo-history profiles, and reranks candidates using profile, prompt, or hybrid scoring. The paper separates two questions that are often conflated in prototypes: whether noisy public traces can be reconstructed well enough to serve as historical movement evidence, and whether profile-aware reranking improves over non-personalized baselines. On an OSM public-trace probe, timestamp coverage and GPS-to-route proximity are strong, but strict reconstruction usability fails because route-distance ratio remains high. To move toward a fair public benchmark, we add a Porto taxi trajectory evaluation with OSM driving-route candidates, feature/path oracles, bootstrap confidence intervals, and NASR-style path-overlap metrics. In this 50-query Porto prototype, a vehicle-aware profile ranker improves over shortest-distance and a generic profile on Hit@1, Hit@3, MRR, NDCG@3, and mean feature distance, while path-level metrics show that map matching and candidate generation remain bottlenecks. The contribution is therefore a transparent, reproducible benchmark framework for preference-aware OSM route ranking, not a premature claim that public GPS traces are clean per-user histories.

CCS Concepts

• **Information systems** → **Location based services**; **Recommender systems**; • **Computing methodologies** → *Natural language processing*.

Keywords

personalized routing, OpenStreetMap, GPS traces, route recommendation, pseudo-history profiles

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1 Introduction

Route recommendation is not only a shortest-path problem. A person may prefer a greener, quieter, simpler, better-lit, or more familiar route even when that route is not the minimum-distance option. This makes routing a natural personalization problem: the best route depends on the origin and destination, but also on the request context and the traveler's implicit or explicit preferences. Prior work on personalized route recommendation often assumes large-scale navigation logs, clean historical route choices, or trained models over dense trajectory datasets [2, 3, 6]. Those assumptions are strong for a research prototype that aims to remain transparent, reproducible, and deployable with open geospatial data.

This paper studies a deliberately constrained alternative: can OSM-derived historical movement signals support preference-aware route ranking when clean user histories are unavailable? We use the term *pseudo-history profile* to avoid overclaiming identity or intent. Public GPS traces can indicate where movement has occurred, but they are not necessarily persistent per-user logs. Treating them as clean preferences would make the evaluation look stronger than it is. Instead, the system explicitly measures reconstruction quality, marks weak reconstruction as a limitation, and evaluates route ranking separately from trace reconstruction.

We present a working OSM-based route-ranking system. Given an origin, destination, request context, optional preference text, and optional history records, the backend generates OSM route candidates, extracts interpretable route features, constructs dynamic profile weights from contextual history, and ranks candidates using profile, prompt, or hybrid modes. The implementation is exposed through a FastAPI backend and React interface, preserving the product-facing demonstration that motivated the project. This matters because route recommendation is ultimately an interactive decision aid: the system must produce ranked alternatives and explanations that a user can inspect, not only offline scores.

The paper also reframes the evaluation around validity. A first OSM public-trace experiment demonstrates the end-to-end pipeline for Toronto walking traces, but strict usability fails because route-distance ratio remains too high. We therefore add a public Porto taxi trajectory benchmark to move closer to recognized route-recovery evaluation. The Porto benchmark reconstructs observed OSM driving routes from public taxi trajectories, generates route candidates for the same origin and destination, and reports ranking metrics plus path-overlap metrics with bootstrap confidence intervals. This does not yet reproduce a full external learned baseline such as NASR [6], but it makes the comparison target explicit and measurable.

The contributions are:

- An end-to-end OSM route-ranking architecture that combines candidate generation, interpretable route features, contextual pseudo-history profiles, prompt-based ranking, and a GUI/API workflow.
- A trace-to-profile pipeline that converts public GPS-derived signals into pseudo-history records while reporting reconstruction quality instead of assuming clean user histories.
- A corrected route-candidate evaluation protocol in which generated candidates are ranked against held-out observed routes, avoiding a weaker history-record ranking setup that favored shortest-distance baselines.

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- A Porto public-dataset benchmark with random, shortest-distance, profile, vehicle-profile, feature-oracle, and path-oracle comparisons using Hit@K, MRR, NDCG@3, feature distance, path F1, NDTW, edit distance, and confidence intervals.

The central claim is intentionally modest but useful: profile-aware OSM reranking can be evaluated transparently against non-personalized baselines, and early public-dataset results show promise, while reconstruction and candidate generation remain the main obstacles to a stronger conference claim.

2 Related Work

2.1 Personalized Route Recommendation

Personalized route recommendation has been studied with stronger data assumptions than this work can currently make. Dai et al. use large-scale trajectory data to model personalized route recommendation from historical movement behavior [2]. More recent user-habit work models route sorting from historical navigation behavior and compares against minimum-ETA, LightGBM, and DCN-v2 baselines with inconsistency-oriented metrics [3]. These papers motivate our use of route features and historical profiles, but they assume cleaner user or vehicle histories than OSM public trace signals provide.

Neural search methods frame route recommendation as an origin-destination search problem. NASR integrates neural models with A* search for personalized route recommendation and evaluates route recovery with path-level metrics such as precision, recall, F1, and edit distance [6]. This is the strongest top-tier methodological reference for our next benchmark target because it affects route generation itself rather than only post-hoc reranking. Our current system is simpler: it generates a small set of OSM candidates and reranks them with interpretable profile or prompt signals. The Porto benchmark is designed to move the evaluation closer to NASR-style route recovery without pretending that a full NASR reproduction has already been completed.

2.2 Trajectory Reconstruction and Historical Route Planning

Trajectory reconstruction and popular-route mining are directly relevant because our pseudo-history profiles depend on reconstructed traces. Chen et al. study popular-route construction from uncertain trajectories and use trajectory similarity metrics such as normalized dynamic time warping and maximum distance [1]. Tian et al. propose DRPK for effective route planning from historical trajectories on road networks [5]. These works motivate a strict separation between reconstruction quality and ranking quality. A route ranker can appear to perform well if the reconstructed reference path is poor or if the candidate pool is artificially easy; therefore, we report route-distance ratio, GPS-to-route distance, path F1, NDTW, edit distance, and oracle upper bounds.

2.3 Contextual and Preference-Aware Recommendation

Contextual route recommendation datasets and route-choice studies show that origin-destination routing is affected by context, not only graph distance [4]. Preference-driven travel-planning systems such as Personal Travel Solver use LLM and solver components to infer user preferences and produce personalized travel plans [7]. Our work is narrower. We do not claim full itinerary planning or language-grounded route generation. Instead, natural-language preferences are optional inputs used to rerank already-generated OSM route candidates. When preference text is synthetic, we label it as synthetic and treat prompt/SBERT results as ablations rather than evidence of real stated preferences.

2.4 Positioning

The closest direct future comparison is not the OSM public-trace experiment alone. The defensible comparison path is a same-data public benchmark, most likely Porto, where our candidate reranker and external route-recovery baselines are evaluated with the same path-overlap metrics. Until that reproduction is complete, this paper positions the proposed method as an interpretable, open-data route-ranking framework with transparent diagnostics, rather than as a system that already surpasses learned personalized search methods.

3 Problem Statement

Let a route-ranking request be defined by origin o, destination d, request context c, optional preference text p, and optional historical records H_u for a user or pseudo-user. The system must generate a candidate set $R = \{r_1, \ldots, r_k\}$ and return an ordering that places routes aligned with observed or inferred preferences near the top.

Each route r_i is represented by an interpretable feature vector:

$$x_i = [\text{distance}, \text{major}, \text{walk}, \text{residential}, \text{service}, \text{turns}, \ldots]. \quad (1)$$

For OSM-derived pseudo-history evaluation, the held-out record is not treated as a generated candidate. Instead, it is treated as an observed reference route with feature vector x^*. The oracle generated candidate is:

$$r^* = \arg\min_{r_i \in R} \text{dist}(\hat{x}_i, \hat{x}^*), \quad (2)$$

where $\hat{x}$ denotes min-max normalized features within the evaluation set.

The research question is:

> Can dynamic profiles built from OSM-derived historical movement signals rank generated OSM route candidates closer to observed pseudo-route behavior than non-personalized baselines?

This is intentionally narrower than claiming full personalized navigation. The paper evaluates a reranking problem: candidate routes are already generated by OSM graph search, and the proposed profile module changes their order. This distinction matters because a learned search method such as NASR can generate different paths, while our current implementation can only choose among candidates in the generated pool.

3.1 Evaluation Assumptions

The evaluation uses two kinds of observed reference routes. In the OSM public-trace setting, references are reconstructed walking-route pseudo-histories from public GPS trackpoints. In the Porto

Table 1: Baseline alignment. The table separates conceptual relevance from direct comparability so that the evaluation does not force an invalid comparison.

Work	Main task	Data signal	Comparable metrics	Role in this paper
User-habit route recommendation [3]	Personalized vehicle route sorting	Historical navigation logs	Inconsistency-oriented ranking metrics	Closest conceptual profile-ranking reference, but not directly comparable without clean user logs.
NASR [6]	Personalized route search/recovery	Trajectory data and road graphs	Precision, recall, F1, edit distance	Main future technical benchmark target on public trajectory data.
Big trajectory PRR [2]	Driver-specific route recommendation	Large GPS trajectory data	Route-cost and preference-oriented evaluation	Older but important driver-preference baseline family.
RICK [1]	Popular route inference from uncertain trajectories	Sparse/noisy trajectories	NDTW, maximum distance	Reconstruction-quality reference, not a personalized ranking baseline.
DRPK [5]	Realistic route planning from historical trajectories	Large historical trajectories	Path prediction and efficiency metrics	Non-personalized historical-trajectory planning reference.
Context Trails / PTS [4, 7]	Contextual route/travel recommendation	Contextual POI trails or reviews	Recommendation and preference-satisfaction metrics	Motivates context and preference modeling; task differs from OSM route reranking.

setting, references are reconstructed driving routes from public taxi trajectories. In both cases, the observed reference is imperfect. We therefore report reconstruction diagnostics and oracle upper bounds rather than treating the held-out route as unquestioned ground truth.

3.2 Baseline Objective

For each held-out query, a method returns an ordering of generated candidates. A successful method should rank the oracle candidate near the top, where the oracle is either the candidate closest in normalized route-feature space or, for path diagnostics, the candidate with strongest overlap with the reconstructed reference path. The main deployable baselines are random, shortest-distance, profile, and vehicle-profile ranking. Feature and path oracles are diagnostic upper bounds, not usable methods.

4 Methodology

The prototype contains six main stages: trace processing, approximate map matching, feature extraction, profile construction, candidate generation, and route ranking. Figure 1 summarizes the data flow.

4.1 OSM Trace Processing

The pipeline probes OSM public GPS trackpoints in a Toronto bounding box, parses GPX files, and builds temporal/spatial pseudo-segments. Segments split when timestamp gaps or spatial jumps exceed configured thresholds. Useful segments are retained when they have sufficient distance and plausible speed. These segments are explicitly treated as OSM-derived historical movement signals, not clean per-user histories.

4.2 Approximate Map Matching

For each useful pseudo-segment, the system builds a local OSM walking graph, maps GPS points to nearest graph nodes, deduplicates consecutive repeated nodes, and connects successive matched nodes with shortest paths. This approximate map matching is intentionally simple and therefore reported with diagnostics: matched-point coverage, pair success rate, GPS-to-route distance, and route-distance ratio.

For Porto taxi trajectories, the evaluator uses an OSM driving graph and simplifies dense GPS traces into spatially separated anchors before shortest-path stitching. The anchor spacing parameter reduces redundant point-to-point stitching when consecutive samples fall on nearby road nodes. In the current benchmark, an 80 meter minimum anchor spacing and 60-anchor cap modestly reduce route-distance-ratio inflation. This is a reconstruction improvement, but not a full probabilistic map matcher.

4.3 Route Feature Extraction

For both reconstructed pseudo-history routes and generated candidate routes, the system extracts OSM-derived features: route distance, major-road percentage, walking-way percentage, residential percentage, service-road percentage, intersections, turns, park proximity, lighting percentage, traffic signals, crossings, tunnel length, and an OSM-only safety proxy. The safety proxy is not a crime model; it is a route-attribute score based on available OSM tags.

OSM public GPS signals / Porto taxi trajectories → temporal-spatial segmentation → approximate map matching → route feature extraction → dynamic profile construction → OSM candidate generation → ranking and evaluation

Figure 1: End-to-end framework. Public movement signals are converted into reconstructed route references and pseudo-history profiles. For a new origin-destination query, the system generates OSM candidates and ranks them with baseline, profile, prompt, or hybrid methods.

4.4 Dynamic Profile Construction

Given a request timestamp, the profile module computes temporal context: weekday/weekend, time bucket, season, and rush-hour status. It selects history records with similar context, averages route features, and converts those averages into feature weights. Context rules then adjust weights for rush hour, evening/night, weekend, summer, and winter.

For the Porto benchmark, the generic walking profile is supplemented with a vehicle-aware profile. This profile uses earlier Porto trips as contextual history and emphasizes driving-relevant features: distance, major-road exposure, residential/service-road share, turns, intersections, traffic signals, and tunnel length. The method remains unsupervised: it does not train on held-out labels, and it does not use the oracle candidate during scoring.

4.5 Candidate Generation

The route generator constructs a local OSM graph around the query and returns a small set of route alternatives. The walking pipeline uses OSM walking networks and route attributes such as footways, crossings, lighting, and park proximity. The Porto pipeline uses an OSM driving graph and generates diverse vehicle candidates by varying edge weights, including length-oriented, major-road-preferring, major-road-avoiding, turn-simple, and local-road-avoiding variants. Keeping the candidate pool small preserves runtime for the GUI/API workflow, but it also creates an upper-bound limitation: no ranker can recover a path that candidate generation never proposes.

4.6 Ranking Modes

The prototype supports five evaluation methods:

- **Random:** random permutation of generated route candidates.
- **Shortest-distance:** candidates sorted by OSM route length.
- **Profile:** candidates scored using dynamic profile weights.
- **Prompt/SBERT:** route summaries ranked by semantic similarity to preference text.
- **Hybrid:** 0.75 profile score plus 0.25 prompt/SBERT score, matching the backend.

Prompt/SBERT and hybrid modes are skipped when no preference text exists. This is the current case for the OSM pseudo-history records.

4.7 Scoring

Profile scoring compares each candidate feature vector with a contextual target profile. For numeric feature j, the score rewards closeness between candidate value x_{ij} and target value μ_j under profile weight w_j:

$$s_{\text{profile}}(r_i) = -\sum_j w_j \cdot |\hat{x}_{ij} - \hat{\mu}_j|. \tag{3}$$

The prompt/SBERT mode converts each route to a textual summary and scores semantic similarity against preference text. The hybrid mode combines normalized profile and prompt scores:

$$s_{\text{hybrid}}(r_i) = 0.75\, s_{\text{profile}}(r_i) + 0.25\, s_{\text{prompt}}(r_i). \tag{4}$$

These weights match the backend implementation and are treated as fixed prototype settings, not tuned hyperparameters.

5 Evaluation

The evaluation has three layers: reconstruction quality, threshold sensitivity, and route-candidate ranking quality. This separation is intentional because public OSM GPS trackpoints are not clean per-user histories.

5.1 Reconstruction Quality

The trace pipeline reports timestamp coverage, pseudo-segments found, useful segment rate, map-match success rate, GPS-to-route distance, and route-distance ratio. Strict usability requires timestamp coverage at least 0.80, at least five useful segments, map-match success rate at least 0.70, median GPS-to-route distance at most 100 meters, and median route-distance ratio in $[0.5, 2.0]$. Exploratory usability relaxes only the route-distance ratio ceiling to 2.5.

5.2 Threshold Sensitivity

To avoid tuning a single threshold setting, we evaluate five segmentation/filtering configurations: the current configuration, a stricter time-gap split, a stricter spatial-jump split, a stricter minimum-distance filter, and a lower speed ceiling. Each configuration is evaluated with the same reconstruction diagnostics. This study tests whether the OSM-derived pseudo-history pipeline is stable under reasonable threshold changes.

5.3 Route-Candidate Baselines

The route-candidate evaluation uses generated route candidates. For each held-out pseudo-history record, the evaluator:

(1) Uses earlier records as profile history.
(2) Uses the held-out origin and destination.
(3) Generates up to $k = 5$ OSM walking-route candidates for the OSM trace setting, or up to $k = 8$ OSM driving candidates for the Porto setting.
(4) Treats the held-out reconstructed route features and coordinates as the observed reference.
(5) Selects the oracle candidate by minimum normalized feature distance to the observed reference.
(6) Ranks candidates using each available baseline.

This replaces a weaker sanity check in which the held-out history record was ranked against previous history records. That older setup made shortest-distance look artificially strong because the held-out pseudo-route could be the shortest candidate by construction.

5.4 Baselines

We compare random ranking, shortest-distance ranking, profile ranking, prompt/SBERT ranking, and hybrid ranking. For Porto, we additionally report a vehicle-aware profile ranker, a feature-oracle upper bound, and a path-oracle upper bound. The vehicle-aware profile ranker uses earlier Porto trip features but emphasizes driving-relevant signals such as distance, road-class mix, turns, intersections, and traffic signals. The feature oracle ranks by the same feature distance used to select the ranking oracle. The path oracle ranks by path overlap with the reconstructed route. Both oracle rows are used only to diagnose candidate-pool quality. Prompt/SBERT and hybrid use synthetic preference text in the OSM pseudo-history experiment; this text is labeled as synthetic and is used only as an ablation, not as evidence of real stated user preferences.

5.5 Metrics

Ranking quality is measured with Hit@1, Hit@3, MRR, NDCG@3, and mean feature distance. Hit@K is one when the oracle candidate appears in the first K positions. MRR is the reciprocal rank of the oracle candidate averaged across queries. NDCG@3 discounts lower-ranked relevant candidates and gives a graded top-three ranking view. Mean feature distance reports how close the top-ranked candidate is to the held-out reference in normalized feature space.

Path similarity is measured with point-overlap precision/recall/F1, normalized dynamic time warping (NDTW), coordinate edit distance, and maximum nearest-point distance between the top-ranked route and the held-out reconstructed route. For the Porto public-dataset benchmark, we also report 95% bootstrap confidence intervals over queries.

5.6 External Benchmark Goal

The current OSM pseudo-history evaluation is not a direct comparison against NASR, RICK, or private user-habit systems. To move toward a submission-ready comparison, we add a Porto public-dataset route-candidate benchmark. The Porto setup uses public taxi trajectories, simplifies dense GPS samples into spatially separated reconstruction anchors, reconstructs an observed route on an OSM driving graph, generates OSM route candidates for the same origin and destination, and reports ranking metrics plus path-overlap metrics. This is closer to NASR-style route recovery, but it still requires a larger sample, matched splits, and reproduced external baselines before supporting a direct claim of superiority.

6 Results

This section reports the current prototype outputs. The results support a strengthened research direction, but not yet a claim of superiority over reproduced external systems.

6.1 Trace Reconstruction

The Toronto OSM trace probe currently contains 25,000 parsed trackpoints with timestamp coverage 1.00. The pipeline found 11 pseudo-segments and 11 useful segments, for a useful segment rate of 1.00. Approximate map matching succeeded for all attempted useful segments. The median GPS-to-route median distance is approximately 6.99 meters, indicating close geometric alignment between sampled points and reconstructed routes. However, the median route-distance ratio is approximately 2.37, above the strict target ceiling of 2.0.

Therefore, the current strict prototype usability flag is false. The exploratory usability flag is true under the looser 2.5 route-distance-ratio ceiling. This distinction is central: the data is useful for exploratory pseudo-history experiments, but the reconstruction is not yet clean enough for strong claims about route fidelity.

6.2 Threshold Sensitivity

Table 2 shows that the route-distance-ratio failure is stable across several reasonable threshold choices. This is useful evidence because it shows the reconstruction limitation is not hidden by one chosen configuration.

6.3 Route-Candidate Ranking

The corrected route-candidate baseline evaluation currently runs on 8 held-out pseudo-history queries. Table 3 reports the current ranking metrics.

The profile method improves over shortest-distance on Hit@1, MRR, NDCG@3, path F1, NDTW, edit distance, and maximum path distance in the current run. However, the sample remains small and noisy. These results are best interpreted as evidence that the evaluation pipeline and profile-ranking idea are functioning, not as proof that the method beats external personalized route-recommendation systems.

6.4 Path-Similarity Metrics

Against held-out reconstructed route coordinates, the profile method achieves mean path F1 of 0.630 and mean NDTW of 0.351, compared with 0.601 and 0.344 for shortest-distance. The improvement is directionally promising but not yet publication-grade without a larger public benchmark and confidence intervals.

6.5 Porto Public-Dataset Prototype

To begin moving toward a same-data public benchmark, we also ran a Porto taxi trajectory prototype using the public ECML/PKDD taxi trajectory data. This experiment samples trips from the Porto training file, reconstructs an observed OSM driving route using simplified GPS anchors, generates up to eight diverse OSM driving-route candidates for the same origin and destination, and ranks the candidates using random, shortest-distance, generic profile, and vehicle-aware profile baselines. We also report two oracle upper bounds: a feature oracle, which ranks candidates by the same feature-distance criterion used to define the ranking oracle, and a path oracle, which ranks candidates by path overlap with the reconstructed route. These oracle rows diagnose candidate-pool

Table 2: Threshold sensitivity over OSM-derived pseudo-history reconstruction. All configurations remain exploratory-usable but fail strict usability due to route-distance ratio.

Config	Useful	GPS med.	Ratio med.	Strict
Current	11	6.99	2.37	No
Stricter gap	12	6.72	2.36	No
Stricter jump	11	6.99	2.37	No
Stricter distance	10	7.41	2.36	No
Lower speed ceiling	9	6.32	2.35	No

Table 3: Current route-candidate baseline results over eight held-out OSM pseudo-history queries. Prompt/SBERT and hybrid use synthetic preference text and should be interpreted as ablations.

Method	Hit@1	Hit@3	MRR	NDCG@3	Mean Dist.
Random	0.375	0.750	0.583	0.668	2.622
Shortest-distance	0.250	0.625	0.456	0.690	2.579
Profile	0.375	0.625	0.540	0.727	2.582
Prompt/SBERT	0.250	0.500	0.481	0.673	2.552
Hybrid	0.250	0.500	0.467	0.672	2.613

quality and are not deployable methods. Prompt/SBERT and hybrid methods are skipped because Porto trajectories do not include natural-language route preferences.

In this 50-query run, the vehicle-aware profile improves over shortest-distance and generic profile on Hit@1, Hit@3, MRR, NDCG@3, and mean feature distance. It remains close to random on Hit@3 and does not improve path F1 or NDTW. The simplified-anchor reconstruction reduces mean observed route-distance ratio from the prior dense-stitching value of about 2.82 to 2.69 and the median from 2.17 to 2.04, while reducing the mean number of reconstruction anchors to 62% of raw trace points. This is an improvement, but not a solution: 28 of 50 evaluated queries still have ratio above 2.0, and the path oracle still reaches only 0.590 path F1 and 0.261 NDTW. The important result is therefore not that the proposed method is already superior to external systems, but that the project now has a concrete public-dataset evaluation path with uncertainty estimates and diagnostics.

6.6 Result Summary

The OSM public-trace experiment shows that the full pseudo-history pipeline is operational, but strict reconstruction quality is not yet sufficient. The Porto experiment provides a stronger public-dataset evaluation path: vehicle-aware profile ranking improves over shortest-distance and the generic profile on ranking metrics, while path metrics and oracle ceilings reveal that reconstruction and candidate generation are still the limiting factors. This combination is useful for a submission because it avoids hiding negative evidence while still showing a concrete technical contribution.

7 Discussion

The current prototype is strongest as a research framework: it exposes the full chain from OSM public GPS signals to route reconstruction, feature extraction, profile construction, candidate generation, ranking, and evaluation. The GUI/demo angle remains valuable because it shows how a user would interact with ranked routes and profile explanations.

The main limitation is data validity. OSM public GPS trackpoints cannot be assumed to represent clean, persistent user histories. The current pseudo-history profiles are useful for exploring whether trace-derived movement signals can inform route ranking, but they are not equivalent to logged user route choices. This affects both profile construction and evaluation.

The second limitation is reconstruction quality. GPS-to-route distance is low, but route-distance ratio is high for several segments. The threshold-sensitivity study shows this limitation persists across reasonable segmentation and filtering choices. The Porto benchmark now uses simplified GPS anchors before shortest-path stitching, which modestly reduces route-distance-ratio inflation, but it does not solve the problem. This suggests the approximate nearest-node plus shortest-path stitching method can still overbuild routes. Before submission, the pipeline should improve map matching, segment splitting, or route simplification so strict usability passes without relaxing the route-distance-ratio threshold.

The third limitation is evaluation size and external comparability. The current route-candidate evaluation uses eight held-out queries from one OSM public trace area and 50 sampled Porto queries. This is enough for a transparent prototype and benchmark framework, but not enough for a claim that the method beats prior personalized route-recommendation systems.

The main path to a strong paper is a same-data public benchmark. NASR-style evaluation on a public dataset such as Porto taxi trajectories would let us compare route-recovery or route-ranking methods with recognized metrics such as precision, recall, F1, and edit distance. A direct claim of superiority should only be made after matching the dataset, task definition, train/test split, and metrics of an external baseline or a faithful reproduction.

The Porto prototype is now useful mainly as a diagnostic. It shows that the vehicle-aware profile scorer can be evaluated on

Table 4: Prototype Porto public-dataset route-candidate results over 50 sampled queries. Values in parentheses are 95% bootstrap confidence intervals. This is a scaled engineering milestone, not a final external-baseline result.

Method	Hit@1	Hit@3	MRR
Feature oracle	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)
Path oracle	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)
Random	0.180 (0.080, 0.281)	0.620 (0.480, 0.760)	0.441 (0.367, 0.523)
Shortest-distance	0.100 (0.020, 0.200)	0.380 (0.240, 0.520)	0.310 (0.245, 0.380)
Profile	0.120 (0.040, 0.220)	0.400 (0.260, 0.540)	0.342 (0.273, 0.417)
Vehicle profile	0.260 (0.140, 0.400)	0.600 (0.440, 0.720)	0.478 (0.393, 0.569)

Method	NDCG@3	Mean Dist.	Path F1	NDTW
Feature oracle	1.000 (1.000, 1.000)	2.104 (1.999, 2.213)	0.541 (0.466, 0.611)	0.236 (0.162, 0.320)
Path oracle	1.000 (1.000, 1.000)	2.307 (2.201, 2.418)	0.590 (0.515, 0.657)	0.261 (0.184, 0.343)
Random	0.605 (0.538, 0.674)	2.317 (2.204, 2.428)	0.514 (0.442, 0.586)	0.225 (0.159, 0.306)
Shortest-distance	0.509 (0.419, 0.604)	2.385 (2.280, 2.480)	0.515 (0.445, 0.586)	0.215 (0.139, 0.289)
Profile	0.517 (0.419, 0.613)	2.374 (2.291, 2.459)	0.451 (0.384, 0.516)	0.169 (0.110, 0.238)
Vehicle profile	0.732 (0.649, 0.808)	2.226 (2.121, 2.333)	0.501 (0.435, 0.566)	0.196 (0.132, 0.271)

public trajectory data and improves over shortest-distance and the generic profile on several ranking metrics. The diverse candidate generator also improves the path-oracle ceiling relative to the earlier five-candidate setup. However, random ranking remains close on some top-k and path metrics, and the oracle rows still show that path-level quality is constrained by the generated candidate pool and reconstructed route references. This points to the next research problem: the candidate-generation/oracle setup and map matching must be strengthened, and the benchmark must be expanded before it can support a strong publication claim.

Prompt/SBERT and hybrid results currently use synthetic preference text generated from reconstructed route features. These are useful as ablations because they test the prompt-ranking machinery, but they do not prove performance with real user-stated preferences. If prompt or hybrid personalization becomes a central claim, the system needs collected preference text or a controlled preference study.

7.1 Threats to Validity

The most important internal-validity threat is oracle definition. Feature-oracle ranking rewards closeness to reconstructed route features, which may favor methods that match aggregate properties rather than exact geometry. Path-oracle metrics partly address this, but they remain bounded by reconstruction quality. The most important external-validity threat is that walking-style OSM pseudo-histories and Porto taxi trajectories represent different travel modes. We therefore do not combine them into one headline result. The most important construct-validity threat is interpreting pseudo-history profiles as user preferences. The paper avoids that interpretation unless clean user identity or stated preferences are available.

7.2 Path to a Stronger Claim

The next publishable claim should be tested on a larger same-data benchmark. A defensible target is: given public Porto trajectories and OSM road networks, compare profile-aware candidate reranking against shortest-path, random, learned feature ranking, and a reproduced NASR-style route-recovery baseline using the same path-overlap metrics. If the profile-aware method improves ranking metrics while maintaining competitive path F1/NDTW and the GUI/API remains usable, the project can claim an interpretable, open-data route-ranking alternative. If it does not, the benchmark will still identify whether the limiting factor is profile scoring, candidate generation, or map matching.

8 Reproducibility

The implementation is organized as a FastAPI backend and React frontend. The backend exposes route ranking through the same workflow used by the evaluation scripts: graph construction, route generation, feature extraction, profile construction, ranking, and result serialization. The OSM pseudo-history files, quality reports, baseline outputs, and Porto benchmark outputs are written under the project data directory. The public Porto CSV is intentionally not committed because of its size; the evaluator writes a placeholder report if the dataset is missing.

The main OSM route-candidate evaluation can be reproduced with the route-candidate baseline script. The public Porto benchmark can be reproduced after placing the Porto training CSV at `data/porto/train.csv`. The 50-query run reported in this paper uses:

```
python scripts\evaluate_porto_candidate_baselines.py
  --max-tests 50 --max-rows-to-scan 12000
  --sample-stride 50 --k-routes 8
  --dist-meters 2500 --bootstrap-samples 1000
  --min-anchor-spacing-m 80 --max-anchors 60
```

The script outputs both JSON and CSV summaries, including skipped-baseline reasons. Prompt/SBERT and hybrid are skipped for Porto because the public trajectories do not contain natural-language preferences. This behavior is deliberate: missing data should produce a reported skip, not fabricated preference labels.

9 Conclusion

This paper presented a dynamic OSM-based route-ranking system that builds pseudo-history profiles from public GPS-derived movement signals, extracts interpretable OSM route features, and ranks generated route candidates using profile, prompt, or hybrid modes. The main methodological point is separation: reconstruction quality, candidate-generation quality, and ranking quality are evaluated independently so that noisy public traces are not mistaken for clean user histories.

The current results show both promise and limits. In the OSM public-trace setting, the pipeline produces usable exploratory pseudo-history signals but fails strict reconstruction usability because route-distance ratio remains high. In the Porto public-dataset benchmark, a vehicle-aware profile ranker improves over shortest-distance and a generic profile on several ranking metrics, while path metrics show that map matching and candidate generation remain bottlenecks. The result is a submission-ready research framework: transparent, reproducible, and honest about what is proven. The next step is a larger Porto benchmark with a reproduced external baseline and stronger map matching, which would allow a direct claim about whether interpretable profile-aware reranking can compete with learned personalized route-search methods.

References

[1] Chen Chen, Daqing Zhang, Bin Guo, Xiaojuan Ma, Gang Pan, and Zhaohui Wu. 2012. Constructing Popular Routes from Uncertain Trajectories. In *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Association for Computing Machinery, New York, NY, USA, 195–203. doi:10.1145/2339530.2339562

[2] Jian Dai, Bin Yang, Chenjuan Guo, and Zhiming Ding. 2015. Personalized Route Recommendation Using Big Trajectory Data. In *Proceedings of the 31st IEEE International Conference on Data Engineering*. IEEE, Seoul, Republic of Korea, 543–554. doi:10.1109/ICDE.2015.7113313

[3] Yinuo Huang, Xin Jin, Miao Fan, Xunwei Yang, and Fangliang Jiang. 2024. Personalized Route Recommendation Based on User Habits for Vehicle Navigation. arXiv:2409.14047 [cs.IR] Accepted by the 2024 4th International Conference on Intelligent Digital Twin Systems.

[4] Pablo Sánchez, Alejandro Bellogín, and José L. Jorro-Aragoneses. 2025. Context Trails: A Dataset to Study Contextual and Route Recommendation. In *Proceedings of the 19th ACM Conference on Recommender Systems*. Association for Computing Machinery, New York, NY, USA, 716–725. doi:10.1145/3705328.3748151

[5] Wei Tian, Jieming Shi, Siqiang Luo, Hui Li, Xike Xie, and Yuanhang Zou. 2023. Effective and Efficient Route Planning Using Historical Trajectories on Road Networks. *Proceedings of the VLDB Endowment* 16, 10 (2023), 2512–2524. doi:10.14778/3603581.3603591

[6] Jingyuan Wang, Ning Wu, Wayne Xin Zhao, Fanzhang Peng, and Xin Lin. 2019. Empowering A* Search Algorithms with Neural Networks for Personalized Route Recommendation. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Association for Computing Machinery, New York, NY, USA, 539–547. doi:10.1145/3292500.3330824

[7] Zihan Wang, Zhiwei Deng, Jiaying Yuan, Tianyang Wang, Hongcheng Guo, Yong Guo, and Zhaopeng Hu. 2025. Personal Travel Solver: A Preference-Driven LLM-Solver System for Travel Planning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Vienna, Austria, 27622–27642. doi:10.18653/v1/2025.acl-long.1339